

Task 2 — Observability Deep-Dive, SLOs & Error Budgets

Data Analyst · Sprint A - Scale & Reliability · Task Brief

DATA ANALYST

PlaceMux · Phase 3 · Task Brief

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Advanced · Advanced

Sprint A - Scale & Reliability · Mastery 62%

Focus

Define the metric layer properly: a single source of truth, with SLAs on freshness and quality.

What to build / complete

- A semantic/metric layer with one definition per metric
- Data-quality tests (freshness, volume, nulls, uniqueness) with alerts
- A published metric dictionary everyone uses

Execution — work through in order

Phase 3 gives you a goal and a bar, not a recipe. These stages are the discipline; the design decisions are yours to make and defend.

Stage A — Understand & set the bar

1. Re-read the Expected Output and the scoring parameters so you build to the bar, not past it.
2. Confirm every prerequisite and upstream input is actually in your hands, not merely promised.
3. Restate, in one sentence, what 'good' means here: Two people asking the same question get the same number - always.

Stage B — Build: A semantic/metric layer with one definition per metric

1. Define the metric, its source & the decision

Define precisely what “A semantic/metric layer with one definition per metric” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage C — Build: Data-quality tests (freshness, volume, nulls, uniqueness) with alerts

1. Define the metric, its source & the decision

Define precisely what “Data-quality tests (freshness, volume, nulls, uniqueness) with alerts” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage D — Build: A published metric dictionary everyone uses

1. Define the metric, its source & the decision

Define precisely what “A published metric dictionary everyone uses” measures, which raw events it depends on, and which decision it changes. No decision, no metric.

2. Build the query/model on real data

Build the query/model on real production data — sampled if you must, but never synthetic.

3. Validate: reconcile, sanity-check, quantify uncertainty

Reconcile against a source system, sanity-check the magnitude, and state the uncertainty honestly.

4. Tie every number to an action & defend it

For every number, be ready to say where it came from and what action it triggers. Undefendable numbers do not ship.

Stage E — Integrate, verify & make demoable

1. Wire your work into the end-to-end flow and run the whole journey once, start to finish.
2. Show the same metric identical across the dashboard, the exec deck and an ad-hoc query.
3. Break it on purpose: force the failure path and confirm it degrades the way you designed.
4. Prepare a live demo on real data. Not slides, not screenshots, not 'it works on my machine'.

Done when (demoable by end of task)

- “A semantic/metric layer with one definition per metric” is complete, real, and demoable end-to-end.
- “Data-quality tests (freshness, volume, nulls, uniqueness) with alerts” is complete, real, and demoable end-to-end.
- “A published metric dictionary everyone uses” is complete, real, and demoable end-to-end.
- Good looks like: Two people asking the same question get the same number - always.

Worry if any of these are true

- The same metric differing across surfaces.
- No freshness or quality alerting.
- Calling an experiment early because it looked good on day two.

Hand-off

- Metric layer to every role. Make sure the next person can pick it up without you.